

TECHNICAL BULLETIN**DIXIE CHEMICAL**

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**EPOXY CURING AGENT 106
(ECA 106)****Typical Properties**

Properties	Value	Test Method
Molecular Weight (average)	167-171	
Specific gravity @ 25°C, g/cc	1.205 ± 0.010	D-102 (ASTM D 4052)
Appearance	Yellow to amber liquid	D-104 (Visual)
Viscosity @ 25 °C, cPs	75-300	D-150F
Gel Time, min	24-33	D-160J
Color, Gardner	8 max	D-150G

Applications

ECA 106 is a low viscosity liquid anhydride epoxy curing agent based on methyltetrahydrophthalic anhydride. For convenience, it is pre-catalyzed with an accelerator that contributes to good glass transition temperature (T_g) in cured polymers. It is designed for use in converting epoxy resins to highly crosslinked polymers with excellent physical and electrical properties. ECA 106 is used in demanding applications including laminating and filament winding. It can also be used in electrical applications such as casting, potting and encapsulation.

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ECA 106K

ECA 106 imparts ease of handling due to its low viscosity, low volatility and low freezing point. It is a liquid at room temperature. If crystallization does occur during shipping or storage, it can be returned to liquid by heating at 38-55°C (100-130°F) for 2-8 hours. Formulations of ECA 106 and liquid epoxies have a long pot life/working time in excess of 24 hours.

For more information on the use of anhydrides like ECA 106 as epoxy curing agents, please consult the Technical Bulletin, FORMULATING ANHYDRIDE-CURED EPOXY SYSTEMS, available from Dixie Chemical.

Formulation Example

ECA 106 was formulated with a variety of epoxy resins. Viscosity was measured at 25°C, and gel time was measured at 100°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

Epoxy type	ECA 106, phr	Viscosity @ 25 °C, cP	Gel Time ¹ , min	Tg ² , °C
Standard BPA Liquid Epoxy	89	840	28	131
Low Viscosity BPA Liquid	92	640	27	138
Cycloaliphatic Epoxy	122	183	30	199
Epoxy Phenol Novolac	96	800	26	138
Epoxy BPA Novolac	87	2950	25	150
BPF Liquid Epoxy	96	500	27	133

1. Shyodu Hot Pot (100g @ 100°C)

2. Cured 1 hr @ 120°C and 1 hr @ 220°C. Analyzed in DSC: 40°C/min to 250°C

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Handling & Storage

ECA 106 will react with water to form diacids. This is normally undesirable, so ECA 106 should be stored in such a way that it is carefully protected from moisture contamination. When properly stored, ECA 106 has a shelf life of 24 months.

For more details on the design of bulk storage for ECA 106, consult the Dixie Chemical brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

ECA 106 is available from Dixie Chemical plant in Pasadena, Texas. Contact your Dixie representative for details.